

UNIT I

1. If 1 and 2 are Eigen values of a 2×2 matrix A, what are the eigen values of A^2 and A^{-1} .

Solution: Eigen values of A^2 are 1 and 4.

Eigen values of A^{-1} are 1 and $\frac{1}{2}$.

2. Given $A = \begin{pmatrix} -1 & 0 & 0 \\ 2 & -3 & 0 \\ 1 & 4 & 2 \end{pmatrix}$, find the Eigen values A^2 .

Solution: A is a triangular matrix. Therefore its Eigen values are the diagonal elements, i.e, the Eigen values of A are -1, -3 and 2. Hence the Eigen values of A^2 are 1, 9 and 4.

3. If the sum of two Eigen values and trace of 3×3 matrix A are equal, find the value of $|A|$

Solution: Let $\lambda_1, \lambda_2, \lambda_3$ be the Eigen values of A. Then we have $\lambda_1 + \lambda_2 = \text{trace of A}$

$\Rightarrow \lambda_1 + \lambda_2 = \lambda_1 + \lambda_2 + \lambda_3 = 0$. Hence $|A| = \text{product of Eigen values} = \lambda_1 \lambda_2 \lambda_3 = 0$.

4. For a given matrix A of order 3, $|A| = 32$ and two of its eigen values are 8 and 2. Find the sum of Eigen values.

Solution: Let the third Eigen value be λ . Then $(8)(2)(\lambda) = |A| = 32 \Rightarrow \lambda = 1$.

Hence the sum of the Eigen values = $8 + 2 + 1 = 11$.

5. Find the sum and product of the Eigen values of the square matrix $A = \begin{pmatrix} 8 & 1 & 6 \\ 3 & 5 & 7 \\ 4 & 9 & 2 \end{pmatrix}$ using the

properties.

Solution: Sum of the Eigen values = Sum of the main diagonal elements = $8 + 5 + 2 = 15$.

Product of the Eigen values = $\text{Det}(A) = -360$.

6. Find the Eigen values of A^1 of the matrix $A = \begin{pmatrix} 2 & 5 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{pmatrix}$.

Solution: A is a triangular matrix. Therefore the Eigen vales of A are 2, 3 and 4 and hence the

Eigen values of A^1 are λ_2, λ_3 and λ_4 .

7. Find the sum of the Eigen values of $2A$ if $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$.

Solution: if $\lambda_1, \lambda_2, \lambda_3$ are the Eigen values of , then $\lambda_1 + \lambda_2 + \lambda_3 = 18$. We know that

$2\lambda_1, 2\lambda_2, 2\lambda_3$ are the Eigen values of $2A$. Therefore the sum of the eigen values of

$$2A = 2(\lambda_1 + \lambda_2 + \lambda_3) = 2(18) = 36.$$

8. Find the eigen values of $2A^2$ if $A = \begin{pmatrix} 4 & 1 \\ 3 & 6 \end{pmatrix}$

Solution: Eigen values of A are given by

$$|A - \lambda I| = 0 \Rightarrow \lambda^2 - 6\lambda + 5 = 0 \Rightarrow \lambda = 1, 5.$$

Hence the Eigen values of $2A^2$ are $2(1)^2, 2(5)^2$. i.e, 2, 50.

9. Find the sum of the squares of the Eigen values of $A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$

Solution: A is a triangular matrix. Therefore the Eigen values of A are 3, 2 and 5. The sum of

squares of the Eigen values of $A^2 = 3^2 + 2^2 + 5^2 = 9 + 4 + 25 = 38$.

10. Find the Eigen values $2A - I$, given $A = \begin{pmatrix} -4 & 1 \\ 3 & -2 \end{pmatrix}$

Solution:

$$2A - I = \begin{pmatrix} -8 & 2 \\ 6 & -4 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -9 & 2 \\ 6 & -5 \end{pmatrix}$$

The Characteristic equation of $2A - I$ is given by

$$\begin{aligned}
 |2A - I - \lambda I| = 0 &\Rightarrow \begin{vmatrix} -9 - \lambda & 2 \\ 6 & -5 - \lambda \end{vmatrix} = 0 \\
 &\Rightarrow \lambda^2 + 14\lambda + 33 = (\lambda + 11)(\lambda + 3) = 0 \\
 &\Rightarrow \lambda = -3, -11
 \end{aligned}$$

11. For what values of 'c' the Eigen values of the matrix $A = \begin{pmatrix} 1 & 2 \\ c & 7 \end{pmatrix}$ are real and unequal, real and equal, complex conjugate?

Solution:

$$\begin{aligned}
 A - \lambda I &= \begin{pmatrix} 1 & 2 \\ c & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{vmatrix} 1 - \lambda & 2 \\ c & 4 - \lambda \end{vmatrix} \\
 &\Rightarrow |A - \lambda I| = (1 - \lambda)(4 - \lambda) - 2c = 0 \\
 &\Rightarrow \lambda^2 - 5\lambda + (4 - 2c) = 0
 \end{aligned}$$

Hence the eigen values are real and unequal, real and equal, complex conjugates according to

$\Delta = b^2 - 4ac$ is $>$, $=$, $<$ zero.

i.e., $(-5)^2 - 4(1)(4 - 2c) > \text{or } = \text{or } < 0$ i.e., $25 - 16 + 8c > \text{or } = \text{or } < 0$

i.e., $c > \frac{-9}{8}$, $c = \frac{9}{8}$, $c < -\frac{9}{8}$.

12. Prove that A and A^T have the same eigen values

Solution: $|A^T - \lambda I| = |A^T - (\lambda I)^T| = |(A - \lambda I)^T| = |A - \lambda I| \Rightarrow A$ and A^T have the same characteristic equation and hence they have the same eigen values.

13. Similar matrices have the same characteristic roots – Prove

Solution: Let A and B two similar matrices, then there exists a matrix P such that $B = P^{-1}AP$.

$$\begin{aligned}
 |B - \lambda I| &= |P^{-1}AP - P^{-1}\lambda IP| = |P^{-1}||A - \lambda I||P| = |A - \lambda I||PP^{-1}| \\
 \text{Hence} \quad &= |A - \lambda I|
 \end{aligned}$$

i.e., A and B have the same characteristic equation. Therefore, they have the same characteristic roots.

14. Define similar matrices and orthogonal matrices.

Solution: Two matrices A and B are said to be similar, if there exists a matrix P such that B

$$= P^{-1}AP$$

A real matrix B is said to be orthogonal $BB^T = B^T B = I$
if

15. Is the matrix $B = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ orthogonal? Justify.

Solution: $BB^T = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$

Similarly, $B^T B = I$. Hence B is orthogonal.

16. If $D = P^{-1}AP$, then find the matrix A^k , where k is a positive integer and A, D, P are square matrices of same order.

Solution:

$$D = P^{-1}AP \Rightarrow PDP^{-1} = PP^{-1}AP^{-1}P = IAI = A$$

$$\text{Therefore, } A^2 = A.A = PDP^{-1}PDP^{-1} = PDIDP^{-1} = PD^2P^{-1}$$

Proceeding in this manner we have $A^k = PD^kP^{-1}$

17. State Cayley-Hamilton theorem

Solution: Every square matrix satisfies its own characteristic equation.

18. Give two uses of Cayley-Hamilton theorem

Solution:

- i. if A is non-singular we can find A^{-1}
- ii. Higher positive integral powers of A can be computed.

19. Use Cayley-Hamilton theorem to find $A^4 - 4A^3 - 5A^2 + A + 2I$ where $A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

Solution:

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 4\lambda - 5 = 0 \Rightarrow A^2 - 4A - 5I = 0 \text{ (By CH Theorem)}$$

$$\Rightarrow A^2(A^2 - 4A - 5I) = 0 \Rightarrow A^4 - 4A^3 - 5A^2 = 0$$

$$\Rightarrow A^4 - 4A^3 - 5A^2 + A + 2 = 0 + A + 2I = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$$

20. To what type of matrices the diagonalisation by orthogonal transformation is possible?

Solution: Real symmetric matrices.

21. Can $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ be diagonalised? why?

Solution: Yes. Even if the eigen values of A are equal, namely 1, 1, it is possible to find two linearly independent eigen vectors corresponding to the eigen value 1.

22. Find the matrix of the quadratic form $2x^2 + 2y^2 + 3z^2 + 2xy - 4xz - 4yz$

Solution: The required matrix $A = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ -2 & -2 & 3 \end{pmatrix}$

23. Write down the matrix corresponding to the quadratic form $x^2 + y^2 + z^2 + 2zx + 4\sqrt{2}yz$

Solution: The required matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2\sqrt{2} \\ 1 & 2\sqrt{2} & 1 \end{pmatrix}$

$\begin{pmatrix} 2 & 1 & -2 \end{pmatrix}$

24. Write down QF corresponding to matrix $A = \begin{pmatrix} 1 & 2 & -2 \\ -2 & -2 & 3 \end{pmatrix}$

Solution: The QF of the matrix is $2x^2 + 2y^2 + 3z^2 + 2xy - 4yz - 4zx$

25. Define index and signature of a quadratic form. Find the index and signature of the quadratic form

$$x^2 + 2x^2 - 3x^2.$$

Solution: The number (p) of positive terms in the canonical form of QF is called the index of the QF. The number of positive terms minus the number of the negative terms is called the signature of the QF.

Index = 2, Signature = 1.

UNIT II

1. Find grad ϕ at the point $(-1, 2, -1)$ when $\phi = 3x^2y - y^3z^2$

Solution:

$$\begin{aligned}\text{grad } \phi &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (3x^2y - y^3z^2) \\ \nabla \phi &= (6xy\vec{i} + (3x^2 - 3y^3z^2)\vec{j} - 16\vec{k}) \\ (\nabla \phi)_{(-1,2,-1)} &= -12\vec{i} - 9\vec{j} - 16\vec{k}\end{aligned}$$

2. If $\vec{r}$ is the position vector of (x, y, z) with respect to origin, then prove that $\nabla(r^n) = nr^{n-2}\vec{r}$

Solution:

$$\begin{aligned}\nabla(r^n) &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) r^n = \vec{i} (nr^{n-1}) \frac{\partial r}{\partial x} + \vec{j} (nr^{n-1}) \frac{\partial r}{\partial y} + \vec{k} (nr^{n-1}) \frac{\partial r}{\partial z} \\ &= \vec{i} (nr^{n-1}) \frac{x}{r} + \vec{j} (nr^{n-1}) \frac{y}{r} + \vec{k} (nr^{n-1}) \frac{z}{r} \\ &= nr^{n-1} (\vec{x}\vec{i} + \vec{y}\vec{j} + \vec{z}\vec{k}) = nr^{n-2} \vec{r}\end{aligned}$$

3. Find the directional derivative of $\phi = xy + yz + zx$ at $(1, 2, 3)$ along the x-axis.

Solution:

$$\begin{aligned}\nabla \phi &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (xy + yz + zx) \\ &= ((x+y)\vec{i} + (x+z)\vec{j} + (y+x)\vec{k}) \\ (\nabla \phi)_{(1,2,3)} &= 5\vec{i} + 4\vec{j} + 3\vec{k}\end{aligned}$$

Along x-axis then $y=0, z=0$ $\therefore \vec{a} = x\vec{i}$ and $|\vec{a}| = 1$

4. Find the directional derivative $\phi = x^2 + yz + 4xz^2$ at the point $(1, -1, -2)$ then (i) that is maximum (ii) in the direction PQ, where Q is $(3, -3, -2)$

Solution:

$$\text{i. } \nabla \phi = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (x^2yz + 4xz^2)$$

$$= \left((2xyz + 4z^2)\vec{i} + (x^2z)\vec{j} + (x^2y + 8xz)\vec{k} \right)$$

$$(\nabla\phi)_{(1,-2,-1)} = 8\vec{i} - \vec{j} - 10\vec{k}$$

The maximum directional derivative of ϕ at $(1, -2, -1)$ is

$$= \sqrt{64 + 1 + 100}$$

$$= \sqrt{165} \text{ units}$$

ii. Directional Derivative of ϕ in the direction of

$$\overrightarrow{PQ} = \nabla\phi \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|}$$

$$= \frac{16+1+10}{\sqrt{4+1+1}}$$

$$= \frac{27}{\sqrt{6}} \text{ units}$$

5. Find the angle between the normal to the surface $xy = z^2$ at the point $(-2, -2, -2)$ and $(1, 9, -3)$

Solution:

$$\nabla\phi = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (xy - z^2)$$

$$= (y\vec{i} + x\vec{j} - 2z\vec{k})$$

$$= (-2\vec{i} - 2\vec{j} - 4\vec{k})$$

$$(\nabla\phi_1)_{(-2,-2,-2)} = 2\vec{i} - \vec{j} + 2\vec{k}, \quad (\nabla\phi_2)_{(1,9,-3)} = 9\vec{i} - \vec{j} + 6\vec{k}$$

$$\cos\theta = \frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1| |\nabla\phi_2|} = \frac{-44}{\sqrt{24}\sqrt{118}}$$

$$\theta = \cos^{-1} \left(\frac{-11}{\sqrt{177}} \right)$$

6. Find the unit normal to the surface $\phi(x, y, z) = x^3 - xyz + z^3 - 1$, at the point $(1, 1, 1)$

Solution:

$$\nabla\phi = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x^3 - xyz + z^3)$$

$$= ((3x^2 - yz)\vec{i} - xz\vec{j} + (3z^2 - xy)\vec{k})$$

$$\begin{aligned}\nabla \phi &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (x^2 y^2 z^4) \\ &= 2xy^2 z^4 \vec{i} + 2x^2 yz^4 \vec{j} + 4x^2 y^2 z^3 \vec{k}\end{aligned}$$

7. Find the equation of the tangent plane to the surface $2xz^2 - 3xy - 4x = 7$ at $(1, -1, 2)$

Solution:

$$\begin{aligned}\nabla \phi &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (2xz^2 - 3xy - 4x - 7) \\ &= ((2z^2 - 3y - 4)\vec{i} - 3x\vec{j} + 4xz\vec{k}) \\ (\nabla \phi)_{(1, -1, 2)} &= \vec{i} - \vec{j} + 2\vec{k}\end{aligned}$$

Equation of the tangent plane is

$$\begin{aligned}(\vec{r} - \vec{a}) \cdot \nabla \phi &= 7(x - 1) - 3(y + 1) + 8(z - 2) = 0 \\ &= 7x - 3y + 8z = 26\end{aligned}$$

8. In what direction from $(3, 1, -2)$ is the directional derivative of $x^2 y^2 z^4$ maximum?

Solution:

$$\begin{aligned}\nabla \phi &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (x^2 y^2 z^4) \\ &= 2xy^2 z^4 \vec{i} + 2x^2 yz^4 \vec{j} + 4x^2 y^2 z^3 \vec{k}\end{aligned}$$

The maximum directional derivative occurs in the direction of

$$(\nabla \phi)_{(3, 1, -2)} = 96(\vec{i} + 3\vec{j} - 3\vec{k})$$

9. Find the equation of the normal plane to the surface $x^2 + y^2 = z$ at $(2, -1, 5)$

Solution:

$$\nabla^2 \phi$$

The equation of normal is

$$(\vec{r} - \vec{a}) \times \nabla \phi = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x-2 & y+1 & z-5 \\ 4 & -2 & -1 \end{vmatrix} = 0$$

$$[-(y+1) + (z-5)]\vec{i} - [-(x-2) - 4(z-5)]\vec{j} + [-2(x-2) - 4(z-5)]\vec{k} = 0$$

and
$$\begin{aligned} 2(z-5) &= y+1 \\ 4(z-5) &= -(x-2) \end{aligned}$$

$$\therefore \frac{x-2}{4} = \frac{x+1}{-2} = \frac{z-5}{-1}$$

10. If $\vec{r}$ is the position vector of the point (x, y, z) then prove that $\nabla |\vec{r}|^2 = 2\vec{r}$

Solution:

$$\begin{aligned} \nabla |\vec{r}|^2 &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2) \\ &= 2x\vec{i} + 2y\vec{j} + 2z\vec{k} \\ &= 2\vec{r} \end{aligned}$$

11. Find ϕ if $\text{grad} \phi = (y^2 - 2xyz^3)\vec{i} + (3 + 2xy - x^2z^3)\vec{j} + (6z^3 - 3x^2yz^2)\vec{k}$

Solution:

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= (y^2 - 2xyz^3) \Rightarrow \phi = xy^2 - x^2yz^3 + c_1 \\ \frac{\partial \phi}{\partial y} &= (3 + 2xy - x^2z^3) \Rightarrow \phi = 3y + x^2y^2 - x^2z^3 + c_2 \\ \frac{\partial \phi}{\partial z} &= (6z^3 - 3x^2yz^2) \Rightarrow \phi = \frac{6z^4}{4} - x^2yz^3 + c_3 \end{aligned}$$

The solution
$$\phi = xy^2 - x^2yz^3 + 3y + \frac{3z^4}{4} + c$$

12. If $\nabla \phi = (yzi + xzj + xyk)$ then find ϕ

Solution:

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= yz \Rightarrow \phi = xyz + c_1 \\ \frac{\partial \phi}{\partial y} &= xz \Rightarrow \phi = xyz + c_2 \\ \frac{\partial \phi}{\partial z} &= xy \Rightarrow \phi = xyz + c_3 \end{aligned}$$

The solution
$$\phi = xyz + c$$

13. if $\vec{F} + z\vec{i} + x\vec{j} + y\vec{k}$, Prove that the curl $\text{curl } \vec{F} = 0$.

Solution:

$$\text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \vec{i} + \vec{j} + \vec{k}$$

$$\text{curl curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1 & 1 & 1 \end{vmatrix} = 0$$

14. If $\vec{F}$ may be irrotational then Find the values of a, b, c so that

$$(x + y + az)\vec{i} + (bx + 2y - z)\vec{j} + (-x + cy + 2z)\vec{k}$$

Solution:

Given $\vec{F}$ is irrotational then

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + y + az & bx + 2y - z & -x + cy + 2z \end{vmatrix} = 0$$

$$(c+1)\vec{i} - (-1-a)\vec{j} + (b-1)\vec{k} = 0$$

The solution

$$a = -1$$

$$b = 1$$

$$\text{and } c = -1$$

15. If $\vec{F} = 3xyz^2\vec{i} + 2xy^2\vec{j} - x^2yz\vec{k}$ find $\nabla \cdot \vec{F}$ at the point (1, -1, 1)

Solution:

$$\nabla \cdot \vec{F} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (3xyz^2\vec{i} + 2xy^2\vec{j} - x^2yz\vec{k})$$

$$= 3yz^2 + 6xy^2 - x^2y$$

$$(\nabla \cdot \vec{F})_{(1, -1, 1)} = -3 + 6 + 1$$

$$= 4$$

16. If $\nabla \phi$ is a solenoidal vector then prove that ϕ is a solution of Laplace equation

Solution:

$$\begin{aligned}\nabla^2 \phi &= \nabla \cdot (\nabla \phi) \\ &= 0 \text{ (since } \phi \text{ is solenoidal)} \\ \Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} &= 0\end{aligned}$$

ϕ is a solution of Laplace equation.

17. If $\vec{u}$ and $\vec{v}$ are irrotational prove that $\vec{u} \times \vec{v}$ is solenoidal.

Solution:

Given $\vec{u}$ and $\vec{v}$ are irrotational then

$$\nabla \times \vec{u} = 0$$

$$\text{and } \nabla \times \vec{v} = 0$$

w.k.t.

$$\begin{aligned}\nabla (\vec{u} \times \vec{v}) &= \vec{v} \cdot (\nabla \times \vec{u}) - \vec{u} \cdot (\nabla \times \vec{v}) \\ &= \vec{v} \cdot 0 - \vec{u} \cdot 0 \\ &= 0\end{aligned}$$

Thus $\vec{u} \times \vec{v}$ is solenoidal.

18. Prove that $\nabla(r^n) = nr^{n-2}\vec{r}$

Solution:

$$\begin{aligned}\nabla r^n &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) r^n \\ &= nr^{n-1} \frac{x}{r} \vec{i} + nr^{n-1} \frac{y}{r} \vec{j} + nr^{n-1} \frac{z}{r} \vec{k} \\ &= nr^{n-2} \vec{r}\end{aligned}$$

19. For what value of k is the vector $r^k \vec{r}$ is solenoidal

Solution:

$$\begin{aligned}\nabla r^k \vec{r} &= \sum \vec{i} \frac{\partial}{\partial x} (r^k x) \\ &= \sum \vec{i} \left[kx r^{k-1} \frac{\partial r}{\partial x} + r^k \right] = \sum \vec{i} \left[kx r^{k-1} \frac{x}{r} + r^k \right] \\ &= \sum \vec{i} [kx^2 r^{k-2} + r^k] \\ &= 3r^k + kr^{k-2} (x^2 + y^2 + z^2) \\ &= 3r^k + kr^k = r^k (3+k)\end{aligned}$$

If $k = -3$ then $r^k \vec{r}$ is solenoidal

20. Determine $f(r)$ so that the vector $f(r)\vec{r}$ is solenoidal

Solution:

$f(r)\vec{r}$ is solenoidal if $\nabla \cdot (f(r)\vec{r}) = 0$

$$\nabla f(r) \cdot \vec{r} + f(r) \nabla \cdot \vec{r} = 0 \Rightarrow \frac{f'(r)}{r} \vec{r} \cdot \vec{r} + 3f(r) = 0$$

$$rf'(r) + 3f(r) = 0 \Rightarrow \frac{f'(r)}{f(r)} + \frac{3}{r} = 0$$

Integrating on both sides with respect r ,

$$\log f(r) + 3 \log r = \log c$$

$$\log r^3 f(r) = \log c$$

$$\text{Thus, } f(r) = \frac{c}{r^3}$$

21. Prove that $\text{curl} [f(r)\vec{r}] = 0$

Solution:

$$f(r)\vec{r} = f(r)(x\vec{i} + y\vec{j} + z\vec{k})$$

$$\text{Now, } \nabla \times f(r)\vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xf(r) & yf(r) & zf(r) \end{vmatrix} = 0$$

22. If ϕ is solenoidal then find $\nabla^2 \phi$

Solution:

$$\nabla^2 \phi = \nabla(\nabla \phi) = 0 \text{ (since } \phi \text{ is solenoidal)}$$

23. If $\vec{F} = (x+3y)\vec{i} + (y-2z)\vec{j} + (x+\lambda z)\vec{k}$ is solenoidal, Find λ .

Solution:

If $\vec{F}$ is solenoidal, then $\nabla \cdot \vec{F} = 0$

$$\frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+\lambda z) = 0$$

$$\Rightarrow 1+1+\lambda = 0$$

$$\Rightarrow 2+\lambda = 0$$

$$\Rightarrow \lambda = -2$$

24. Find the value of 'a', if $\vec{F} = (2x-5y)\vec{i} + (x+\lambda y)\vec{j} + (3x-z)\vec{k}$ is solenoidal.

Solution:

If $\vec{F}$ is solenoidal, then $\nabla \cdot \vec{F} = 0$

$$\frac{\partial}{\partial x}(2x-5y) + \frac{\partial}{\partial y}(x+\lambda y) + \frac{\partial}{\partial z}(3x-z) = 0$$

$$2+\lambda-1=0 \Rightarrow 1+\lambda=0 \Rightarrow \lambda=-1$$

25. Find the value of 'a', if the vector $\vec{F} = (axy-z^3)\vec{i} + (a-2)x^2\vec{j} + (1-a)xz^2\vec{k}$ is irrotational.

Solution:

If $\vec{F}$ is irrotational, then $\nabla \times \vec{F} = 0$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ axy-z^3 & (a-2)x^2 & (1-a)xz^2 \end{vmatrix} = 0.$$

$$i(0-0) - j((1-a)z^2 + 3z^2) + k(2(a-2) - ax) = 0$$

$$(1-a)z^2 = -3z^2 \Rightarrow 1-a = -3 \Rightarrow a = 4.$$

26. Find the unit vector normal to the surface $x^2 + y^2 - z = 10$ at $(1, 1)$

Solution:

$$\text{Given } \phi = x^2 + y^2 - z - 10$$

$$\nabla \phi = \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} = 2x \vec{i} + 2y \vec{j} + \vec{k}$$

$$\nabla \phi_{(1,1,1)} = 2\vec{i} + 2\vec{j} + \vec{k}$$

$$|\nabla \phi| = \sqrt{4+4+1} = \sqrt{9} = 3$$

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2\vec{i} + 2\vec{j} + \vec{k}}{3}$$

27. If $\vec{F} = x^3 \vec{i} + y^3 \vec{j} + z^3 \vec{k}$. Find $\text{div curl } \vec{F}$.

Solution:

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^3 & y^3 & z^3 \end{vmatrix} = \vec{i}(0-0) - \vec{j}(0-0) + \vec{k}(0-0) = 0.$$

28. Calculate the curl of the vector $\vec{F} = xyz \vec{i} + 3x^2y \vec{j} + (xz^2 - y^2z) \vec{k}$ at $(1, -1, 1)$.

Solution:

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & 3x^2y & xz^2 - y^2z \end{vmatrix} = \vec{i}(-2yz - 10) - \vec{j}(z^2 - xy) + \vec{k}(6xy - xz).$$

$$\left(\nabla \times \vec{F} \right)_{(1,-1,1)} = 2\vec{i} - \vec{j}(1+1) + \vec{k}(-6-1) = 2\vec{i} - 2\vec{j} - 7\vec{k}.$$

29. Find $\text{div } \vec{F}$ where $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$

Solution: Given $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$

$$\vec{F} = (3x^2 - 3yz) \vec{i} + (3y^2 - 3xz) \vec{j} + (3z^2 - 3xy) \vec{k}$$

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(3x^2 - 3yz) + \frac{\partial}{\partial y}(3y^2 - 3xz) + \frac{\partial}{\partial z}(3z^2 - 3xy) \\ &= 6x + 6y + 6z = 6(x + y + z). \end{aligned}$$

30. Find $\text{curl } \vec{F}$ where $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$

Solution: Given $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$

$$\vec{F} = (3x^2 - 3yz) \vec{i} + (3y^2 - 3xz) \vec{j} + (3z^2 - 3xy) \vec{k}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 3yz & 3y^2 - 3xz & 3z^2 - 3xy \end{vmatrix} = \vec{i}(3x - 3x) - \vec{j}(3y - 3y) + \vec{k}(3z - 3z) = 0.$$

31. If $\vec{r} = x \vec{i} + y \vec{j} + z \vec{k}$. Find $\nabla \cdot \vec{r}$ and $\nabla \times \vec{r}$.

$$\text{Solution: } \nabla \cdot \vec{r} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (x \vec{i} + y \vec{j} + z \vec{k}) = 1 + 1 + 1 = 3$$

$$\nabla \times \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0.$$

36. Prove that $\text{grad}(\phi\psi) = \phi \text{ grad } \psi + \psi \text{ grad } \phi$.

Solution:

$$\begin{aligned}\text{grad}(\phi\psi) &= \nabla(\phi\psi) = \sum \vec{i} \frac{\partial}{\partial x}(\phi\psi) \\ &= \sum \vec{i} \left(\phi \frac{\partial \psi}{\partial x} + \psi \frac{\partial \phi}{\partial x} \right) \\ &= \sum \vec{i} \phi \frac{\partial \psi}{\partial x} + \sum \vec{i} \psi \frac{\partial \phi}{\partial x} \\ &= \phi \sum \vec{i} \frac{\partial \psi}{\partial x} + \psi \sum \vec{i} \frac{\partial \phi}{\partial x} \\ &= \phi \nabla \psi + \psi \nabla \phi \\ &= \phi \text{ grad } \psi + \psi \text{ grad } \phi.\end{aligned}$$

37. Find the D.D of $\phi = x^2yz + 4xz^2 + xyz$ at $(1, 2, 3)$ in the direction of the vector $2\vec{i} + \vec{j} - \vec{k}$.

Solution:

$$\text{Given } \phi = x^2yz + 4xz^2 + xyz$$

$$\begin{aligned}\nabla \phi &= \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} \\ &= (2xyz + 4x + yz) \vec{i} + (x^2z + xz) \vec{j} + (x^2y + 8xz + xy) \vec{k}.\end{aligned}$$

$$\nabla \phi_{(1,2,3)} = 54 \vec{i} + 6 \vec{j} + 28 \vec{k} \text{ and}$$

$$\text{given } a = 2 \vec{i} + \vec{j} - \vec{k}.$$

$$\therefore |a| = \sqrt{6}$$

$$D.D = \nabla \phi \frac{a}{|a|}$$

$$= 54 \vec{i} + 6 \vec{j} + 28 \vec{k} \cdot \frac{2 \vec{i} + \vec{j} - \vec{k}}{\sqrt{6}} = \frac{86}{\sqrt{6}}.$$

38. Find the Directional Derivative of xyz at the point $(2, 1, 1)$ in the direction of

$$\vec{j} + \vec{k}.$$

Solution:

$$\text{Given } \phi = xyz$$

$$\nabla \phi = yz \vec{i} + xz \vec{j} + xy \vec{k}.$$

$$a = \nabla \phi_{(2,1,1)} = \vec{i} + 2 \vec{j} + 2 \vec{k}, \quad b = \vec{j} + \vec{k}$$

$$\begin{aligned}\text{D.D in the direction of } \vec{j} + \vec{k} &= \frac{a \cdot b}{|b|} \\ &= \frac{4}{\sqrt{2}}\end{aligned}$$

39. What is the greatest rate of increase of $\phi = xyz^2$ at $(1, 0, 3)$.

Solution:

$$\begin{aligned}\nabla \phi &= \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} \\ &= \vec{i} yz^2 + \vec{j} xz^2 + \vec{k} 2xyz.\end{aligned}$$

$$\nabla \phi_{(1,0,3)} = 0 \vec{i} + 9 \vec{j} + 0 \vec{k}.$$

$$\text{The greatest rate of increase} = |\nabla \phi| = \sqrt{81} = 9 \text{ units}.$$

40. If $\vec{F} = 3xy\vec{i} - y^2\vec{j}$ evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve in the xy plane $y = 2x^2$ from (0,0) to (1,1).

Solution:

$$\begin{aligned} \vec{F} &= 3xy\vec{i} - y^2\vec{j} \\ d\vec{r} &= dx\vec{i} + dy\vec{j} + dz\vec{k} \\ \vec{F} \cdot d\vec{r} &= 3xydx - y^2dy \\ \text{given } y &= 2x^2 \Rightarrow dy = 4xdx \\ \vec{F} \cdot d\vec{r} &= (6x^3 - 16x^5)dx \\ \int_C \vec{F} \cdot d\vec{r} &= \int_0^1 (6x^3 - 16x^5)dx = \frac{7}{6} \end{aligned}$$

41. If $\vec{F} = (4xy - 3x^2z^2)\vec{i} + 2x^2\vec{j} - 2x^3zk$. Check whether the integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path c.

Solution:

$$\begin{aligned} \vec{F} &= (4xy - 3x^2z^2)\vec{i} + 2x^2\vec{j} - 2x^3zk \\ \vec{F} \cdot d\vec{r} &= (4xy - 3x^2z^2)dx + 2x^2dy - 2x^3zdz \\ \int_C \vec{F} \cdot d\vec{r} &= \int_C (4xy - 3x^2z^2)dx + 2x^2dy - 2x^3zdz \\ \text{This integral is independent of the path if } \vec{F} &= \nabla\phi \text{ \& } \nabla \times \vec{F} = 0 \end{aligned}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4xy - 3x^2z^2 & 2x^2 & -2x^3z \end{vmatrix} = 0.$$

Therefore the line integral is independent of the path.

42. If the vector $\vec{F} = x^2\vec{i} + xy^2\vec{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ from (0,0) to (1,1) along the path $y = x$.

Solution: $\vec{F} = x^2\vec{i} + xy^2\vec{j}$ Given $x=y$ $dx=dy$.

$$\begin{aligned} \vec{F} &= x^2\vec{i} + xy^2\vec{j} \quad d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k} \\ \vec{F} \cdot d\vec{r} &= x^2dx + xy^2dy = (x^2 + x^3)dx \\ \int_C \vec{F} \cdot d\vec{r} &= \int_0^1 (x^2 + x^3)dx = \frac{7}{12}. \end{aligned}$$

43. Find $\iiint_S \vec{r} \cdot d\vec{s}$ where s is the surface of the tetrahedron whose vertices are (0,0,0)(0,1,0)(0,0,1)(1,0,0).

$$\text{Solution: } \iiint_S \vec{r} \cdot d\vec{s} = \iiint_V \nabla \cdot \vec{r} \, dv = 3 \iiint_V dv = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dx dy dz = 3.$$

44. If $\vec{F} = 5xy\vec{i} + 2y\vec{j}$ evaluate $\int_C \vec{F} \cdot d\vec{r}$ where c is the part of the curve $y = x^3$

between $x=1$ and $x=2$.

Solution: $\vec{F} \cdot d\vec{r} = 5xydx + 2ydy$

$$\text{given } y = x^3 \Rightarrow dy = 3x^2dx$$

$$\vec{F} \cdot d\vec{r} = 5xydx + 2ydy = 5x^4dx + 6x^5dx$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_1^2 (5x^4 + 6x^5)dx = 46$$

45. Evaluate $\iint_S \vec{a} \cdot \vec{n} ds$ where $\vec{a} = 3x\vec{i} + 4y\vec{j} + 5z\vec{k}$ and S is a closed surface whose volume is V .

Solution: By Divergence theorem

$$\iint_S \vec{F} \cdot \vec{n} ds = \iiint_V \text{div } \vec{F} dV$$

$$\therefore \iint_S \vec{a} \cdot \vec{n} ds = \iiint_V \text{div } \vec{a} dV$$

$$\begin{aligned} \text{div } \vec{a} &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (3x\vec{i} + 4y\vec{j} + 5z\vec{k}) \\ &= 12 \end{aligned}$$

$$\iint_S \vec{a} \cdot \vec{n} ds = \iiint_V 12 dV = 12V.$$

46. Using Green's theorem, find the area of the circle with radius is 'a'.

Solution:

By Green's theorem

$$\text{Area} = \frac{1}{2} \int_C x dy - y dx$$

For a circle with radius r we have $x^2 + y^2 = r^2$

Put $x = r \cos \theta, y = r \sin \theta$

$\therefore dx = -r \sin \theta d\theta, dy = r \cos \theta d\theta$

θ varies from 0 to 2π .

$$\begin{aligned} \text{Area} &= \frac{1}{2} \int_0^{2\pi} r \cos \theta r \cos \theta d\theta - r \sin \theta (-r \sin \theta) d\theta \\ &= \pi r^2 \end{aligned}$$

47. Evaluate $\iint_S \vec{r} \cdot \vec{n} ds$ where S is the closed surface

Solution: By Divergence theorem $\iint_S \vec{F} \cdot \vec{n} ds = \iiint_V \text{div } \vec{F} dV$

$$\therefore \iint_S \vec{r} \cdot \vec{n} ds = \iiint_V \text{div } \vec{r} dV$$

$$= \iiint_V \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (x\vec{i} + y\vec{j} + z\vec{k}) dV$$

$$= \iiint_V 3 dV = 3 \iiint_V dV = 3V.$$

48. Show that $\iint_S \text{curl } \vec{F} \cdot \vec{n} ds = 0$ Where S is any closed surface.

Solution:

We know that Divergence theorem is

$$\iint_S \vec{F} \cdot \vec{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\iint_S \text{curl } \vec{F} \cdot \vec{n} ds = \iiint_V \nabla \cdot (\text{curl } \vec{F}) dv$$

Where V is the volume of the closed surface S .

Since $\nabla \cdot (\text{curl } \vec{F}) = 0$ we get

$$\iiint_V \nabla \cdot (\text{curl } \vec{F}) = 0 \Rightarrow \iint_S \text{curl } \vec{F} \cdot \vec{n} ds = 0$$

49. Find $\nabla \phi$ if $\phi = x^2 + y^2 + z^2$ at $(1, -1, 1)$.

Sol: $\nabla \phi = 2x\vec{i} + 2y\vec{j} + 2z\vec{k}$

$$\nabla \phi_{(1, -1, 1)} = 2\vec{i} - 2\vec{j} + 2\vec{k}$$

50. Using Stokes' theorem prove that $\int_C \vec{r} d\vec{r} = 0$

Solution: By Stokes theorem

$$\int_C \vec{F} d\vec{r} = \iint_S \nabla \times \vec{F} \cdot \hat{n} ds$$

$$\text{Let } \vec{F} = \vec{r} \text{ then } \nabla \times \vec{F} = \nabla \times \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0$$

$$\therefore \int_C \vec{r} d\vec{r} = 0$$

51. Prove that $\nabla r^2 = 2\vec{r}$

Sol:

$$\begin{aligned} \nabla^2 r &= 2r \nabla r & \{ \because \nabla f(r) &= f'(r) \nabla r \} \\ &= 2r \frac{\vec{r}}{r} = 2\vec{r} \end{aligned}$$

UNIT III And UNIT IV

1. Define Laplace Transform of a function $f(t)$

Solution :

Let $f(t)$ be a function of t for $t > 0$, then Laplace transform of $f(t)$ denoted by $L(f(t))$ is defined as

$$L(f(t)) = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

2. State the condition under which the Laplace Transform of $f(t)$ exists.

Solution :

The Laplace transform of $f(t)$ exists, if $f(t)$ is piecewise continuous on $t \geq 0$ and satisfied the condition $|f(t)| \leq M e^{at}$, where M & a are constants

3. State the first and second shifting theorem on Laplace transform

➤ If $L(f(t)) = F(s)$ then $L(e^{at}f(t)) = F(s - a)$

➤ If $L(f(t)) = F(s)$ then $L(e^{-at}f(t)) = F(s + a)$

4. If $L(f(t)) = F(s)$ the $L(f(at)) = \underline{\hspace{2cm}} \left[Ans = \frac{1}{a} F\left(\frac{s}{a}\right) \right]$

5. State the Convolution theorem on Laplace transform

Solution :

If $L(f(t)) = F(s)$, $L(g(t)) = G(s)$ then $L(f(t) * g(t)) = L(f(t)) * L(g(t)) = F(s) * G(s)$

6. Find $L(t^3 e^{-5t})$

Solution :

$$\{L(t^3)\}_{s \leftarrow s+5} = \left\{ \frac{3!}{s^{3+1}} \right\}_{s \leftarrow s+5} = \left\{ \frac{6}{(s+5)^4} \right\}_{s \leftarrow s+5}$$

7. Find $L(\sin 5t \cos 2t)$

$$\text{Solution: } L(\sin 5t \cos 2t) = \frac{1}{2} L(\sin 7t + \sin 3t) = \frac{1}{2} \left(\frac{7}{s^2 + 49} + \frac{3}{s^2 + 9} \right)$$

8. Find $L\left(\frac{\sin t}{t}\right)$

Solution :

$$L\left(\frac{\sin t}{t}\right) = \int_s^{\infty} L(\sin t) ds = \int_s^{\infty} \frac{1}{s^2 + 1} ds = [\tan^{-1} s]_s^{\infty} = \frac{\pi}{2} - \tan^{-1} s$$

9. Find $L\left(\frac{\sin at}{t}\right)$

Solution :

$$\begin{aligned} L\left(\frac{\sin at}{t}\right) &= \int_s^{\infty} L(\sin at) ds = \int_s^{\infty} \frac{a}{s^2 + a^2} ds = [\tan^{-1} s]_s^{\infty} \\ &= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{a}\right) = \cot^{-1}\left(\frac{s}{a}\right) \end{aligned}$$

10. Find $L(\cos 2t \sin 4t)$

Solution:

$$L(\cos 2t \sin 4t) = \frac{1}{2} L(\sin 6t + \sin 2t) = \frac{1}{2} \left(\frac{6}{s^2 + 36} + \frac{2}{s^2 + 4} \right)$$

11. Find $L(\cosh t \sin 2t)$

Solution:

$$\begin{aligned} L\left(\frac{e^t + e^{-t}}{2} \sin 2t\right) &= \frac{1}{2} [L(e^t \sin 2t) + L(e^{-t} \sin 2t)] = \frac{1}{2} \left[\frac{2}{(s-1)^2 + 4} + \frac{2}{(s+1)^2 + 4} \right] \\ &= \frac{1}{(s-1)^2 + 4} + \frac{1}{(s+1)^2 + 4} \end{aligned}$$

12. Find $L(2^t)$

Solution:

$$L(2^t) = L(e^{t \log 2}) = \frac{1}{s - \log 2}$$

13. Can the Laplace transform of $\tan t$ exist? If not give reason

Solution :

The Laplace transform of $\tan t$ does not exist, because $\tan t$ is not piecewise continuous and is not of exponential order.

14. Give the Laplace transform of the Dirac Delta function

Solution : $L[\delta(t-a)] = e^{-as}$

15. Find $L\left(\frac{t}{e^t}\right)$

Solution :

$$L\left(\frac{t}{e^t}\right) = L(e^{-t}t) = [L(t)]_{s \leftarrow s+1} = \left[\frac{1}{s^2}\right]_{s \leftarrow s+1} = \left[\frac{1}{(s+1)^2}\right]$$

16. Find $L(\sqrt{t})$

$$\text{Solution : } L(\sqrt{t}) = \left[\frac{\Gamma\left(\frac{1}{2} + 1\right)}{s^{\frac{1}{2}+1}} \right] = \left[\frac{\frac{1}{2} \Gamma\left(\frac{1}{2}\right)}{s^{\frac{3}{2}}} \right] = \left[\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}} \right]$$

17. Find $L\left(\frac{1}{\sqrt{t}}\right)$

$$\text{Solution : } L\left(\frac{1}{\sqrt{t}}\right) = \left[\frac{\Gamma\left(-\frac{1}{2} + 1\right)}{s^{-\frac{1}{2}+1}} \right] = \left[\frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} \right] = \left[\frac{\sqrt{\pi}}{s^{\frac{1}{2}}} \right]$$

18. Find $L(e^{-t}u(t-1))$

Solution :

$$L(e^{-t}u(t-1)) = [L(u(t-1))]_{s \leftarrow s+1} = \left[\frac{e^{-s}}{s} \right]_{s \leftarrow s+1} = \left[\frac{e^{-(s+1)}}{s+1} \right]$$

19. Find the Laplace of $e^{-2t}t^{1/2}$

Solution :

$$\begin{aligned} L(e^{-2t}t^{1/2}) &= L\left(t^{\frac{1}{2}}\right)_{s \leftarrow s+2} = \left[\frac{\Gamma\left(\frac{1}{2} + 1\right)}{s^{\frac{1}{2}+1}} \right]_{s \leftarrow s+2} = \left[\frac{\frac{1}{2} \Gamma\left(\frac{1}{2}\right)}{s^{\frac{3}{2}}} \right]_{s \leftarrow s+2} \\ &= \left[\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}} \right]_{s \leftarrow s+2} = \left[\frac{\sqrt{\pi}}{2(s+2)^{\frac{3}{2}}} \right] \end{aligned}$$

20. Find $L\left(\frac{e^{-at} - e^{-bt}}{t}\right)$

Solution :

$$\begin{aligned} L\left(\frac{e^{-at} - e^{-bt}}{t}\right) &= \int_s^\infty L(e^{-at} - e^{-bt}) ds = \int_s^\infty \left(\frac{1}{s+a} - \frac{1}{s+b}\right) ds \\ &= [\log(s+a) - \log(s+b)]_s^\infty = \log\left(\frac{s+b}{s+a}\right) \end{aligned}$$

21. Without integrating, by Laplace transform evaluate $\int_0^\infty e^{-4t} \sin 3t dt$

Solution :

$$L\left(\int_0^\infty e^{-4t} \sin 3t dt\right) = [L(\sin 3t)]_{s=4} = \left[\frac{3}{s^2+9}\right]_{s=4} = \left[\frac{3}{25}\right]$$

22. Evaluate $\int_0^\infty te^{-3t} \sin t dt$ using Laplace Transform

Solution :

$$\begin{aligned} \int_0^\infty te^{-3t} \sin t dt &= [L(tsint)]_{s=3} = \left[-\frac{d}{ds}\left(\frac{1}{s^2+1}\right)\right]_{s=3} \\ &= \left[\left(\frac{2s}{(s^2+1)^2}\right)\right]_{s=3} = \frac{3}{50} \end{aligned}$$

23. Find the Laplace transform of $te^{-t} \sin t$

Solution :

$$L(te^{-t} \sin t) = \left[-\frac{d}{ds}\left(\frac{1}{s^2+1}\right)\right]_{s \rightarrow s+1} = \left(\frac{2(s+1)}{(s+1)^2+1}\right)$$

24. If $f(t)$ is a periodic function with period T , what is its Laplace transform?

Solution :

$$L[f(t)] = \frac{1}{1-e^{-Ts}} \int_0^T e^{-st} f(t) dt$$

25. Find $L\left(\frac{1-e^{-2t}}{t}\right)$

Solution :

$$\begin{aligned} L\left(\frac{1-e^{-2t}}{t}\right) &= \int_s^\infty L(1-e^{-2t}) ds = \int_s^\infty \left(\frac{1}{s} - \frac{1}{s+2}\right) ds \\ &= [\log s - \log(s+2)]_s^\infty = \log\left(\frac{s+2}{s}\right) \end{aligned}$$

26. Find $L(t \sin 2t)$

Solution :

$$\begin{aligned} L(t \sin 2t) &= -\frac{d}{ds}[L(\sin 2t)] = -\frac{d}{ds}\left(\frac{2}{s^2+4}\right) \\ &= -\left(\frac{(s^2+4)0 - 2 \cdot 2s}{(s^2+4)^2}\right) = \frac{4s}{(s^2+4)^2} \end{aligned}$$

27. State the convolution theorem on Laplace transform

Solution :

If $L[f(t)] = F(s)$ and $L[g(t)] = G(s)$ then

$$L\left(\int_0^t f(y)g(t-y)dy\right) = F(s)G(s)$$

28. Define the unit step function

Solution :

The unit step function, also called Heaviside's unit function is defined as

$$U(t-a) = \begin{cases} 0, & \text{for } t < a \\ 1, & \text{for } t > a \end{cases} \quad \text{this is the unit step functions at } t = a$$

It is also denoted by $H(t-a)$

29. Give two examples of functions which have no Laplace transform

Solution :

$$e^{t^2} \text{ and } \frac{1}{t^2} \text{ have no Laplace transform}$$

30. State Initial value theorem

Definition :

$$\text{If } L(f(t)) = F(s) \text{ then } \lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s)$$

31. State Final value theorem

Definition :

$$\text{If } L(f(t)) = F(s) \text{ then } \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

32. Find $L^{-1}\left(\frac{s+1}{s^2+2s+2}\right)$

Solution :

$$L^{-1}\left(\frac{s+1}{s^2+2s+2}\right) = L^{-1}\left(\frac{s+1}{(s+1)^2+1}\right) = e^{-t} L^{-1}\left(\frac{s}{s^2+1}\right) = e^{-t} \cos t$$

33. Find $L^{-1}\left(\frac{s}{(s+2)^2}\right)$

Solution :

$$\begin{aligned} L^{-1}\left(\frac{s}{(s+2)^2}\right) &= L^{-1}\left(\frac{s+2-2}{(s+2)^2}\right) = L^{-1}\left(\frac{s+2}{(s+2)^2}\right) - L^{-1}\left(\frac{2}{(s+2)^2}\right) \\ &= e^{-2t} L^{-1}\left(\frac{1}{s}\right) - 2e^{-2t} L^{-1}\left(\frac{1}{s^2}\right) = e^{-2t} (1 - 2t) \end{aligned}$$

34. Find $L^{-1}\left(\frac{s+3}{s^2+4s+13}\right)$

Solution :

$$\begin{aligned} L^{-1}\left(\frac{s+3}{s^2+4s+13}\right) &= L^{-1}\left(\frac{(s+2)+1}{(s+2)^2+9}\right) = e^{-2t} L^{-1}\left(\frac{s+1}{s^2+9}\right) \\ &= e^{-2t} \left\{ L^{-1}\left(\frac{s}{s^2+9}\right) + \frac{1}{3} L^{-1}\left(\frac{1.3}{s^2+3^2}\right) \right\} = e^{-2t} \left(\cos 3t + \frac{1}{3} \sin 3t \right) \end{aligned}$$

35. Find $L^{-1}\left(\frac{s+2}{s^2+4s+5}\right)$

Solution :

$$L^{-1}\left(\frac{s+2}{s^2+4s+5}\right) = L^{-1}\left(\frac{(s+2)}{(s+2)^2+1}\right) = e^{-2t} L^{-1}\left(\frac{s}{s^2+1}\right) = e^{-2t} \sin t$$

36. Find $L^{-1}\left(\frac{s}{4s^2-25}\right)$

Solution :

$$L^{-1}\left(\frac{s}{4s^2-25}\right) = \frac{1}{4} L^{-1}\left(\frac{s}{s^2-\left(\frac{25}{4}\right)}\right) = \frac{1}{4} \cosh \frac{5}{2} t$$

37. Find $L^{-1}\left(\frac{1}{s^2(s^2+1)}\right)$

Solution :

$$L^{-1}\left(\frac{1}{s^2(s^2+1)}\right) = L^{-1}\left(\frac{1}{s^2}\right) - L^{-1}\left(\frac{1}{(s^2+1)}\right) = t - \sin t$$

38. Find $L(t \sin^2 t)$

Solution :

$$\begin{aligned} L(t \sin^2 t) &= -\frac{d}{ds} \left\{ \frac{1}{2} [L(1) - L(\cos 2t)] \right\} = -\frac{d}{ds} \left\{ \frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2+4} \right] \right\} \\ &= \frac{1}{2s^2} - \frac{s^2-4}{2(s^2+4)^2} \end{aligned}$$

39. Find $L^{-1}\left(\frac{1}{s(s+2)}\right)$

Solution : $L^{-1}\left(\frac{1}{s(s+2)}\right) = \frac{1}{2} \left\{ L^{-1}\left(\frac{1}{s}\right) - L^{-1}\left(\frac{1}{(s+2)}\right) \right\} = \frac{1}{2}(1 - e^{-2t})$

40. Find $L^{-1}\left(\log\left(1+\frac{1}{s}\right)\right)$

Solution :

$$\begin{aligned} F(s) &= \log\left(1+\frac{1}{s}\right) = \log(s+1) - \log s \Rightarrow F^1(s) = \frac{1}{s+1} - \frac{1}{s} \\ L^{-1}(F(s)) &= \left(-\frac{1}{t}\right) L^{-1}(F^1(s)) = \left(-\frac{1}{t}\right) L^{-1}\left(\frac{1}{s+1} - \frac{1}{s}\right) \\ &= \left(-\frac{1}{t}\right)(e^{-t} - 1) = \frac{1-e^{-t}}{t} \end{aligned}$$

41. Find $L^{-1}[e^{-2s}]$

Solution : $L^{-1}[e^{-sa}] = \delta(t-a) \Rightarrow L^{-1}[e^{-2s}] = \delta(t-2)$

42. Find $L^{-1}\left(\tan^{-1}\left(\frac{1}{s}\right)\right)$ or $L^{-1}(\cot^{-1}(s))$

Solution :

$$F(s) = \tan^{-1}\left(\frac{1}{s}\right) \Rightarrow F^1(s) = \frac{1}{1+\left(\frac{1}{s}\right)^2} \left(-\frac{1}{s^2}\right) = -\frac{1}{s^2+1}$$

$$L^{-1}(F(s)) = \left(-\frac{1}{t}\right) L^{-1}(F^1(s)) = \left(-\frac{1}{t}\right) L^{-1}\left(-\frac{1}{s^2+1}\right) = \frac{\sin t}{t}$$

43. Find $L^{-1}\left(\frac{s+2}{s^2+4s-12}\right)$

Solution :

$$L^{-1}\left(\frac{s+2}{s^2+4s-12}\right) = L^{-1}\left(\frac{s+2}{(s+2)^2-16}\right) = e^{-2t} L^{-1}\left(\frac{s}{s^2-16}\right) = e^{-2t} \sinh 4t$$

44. Find by convolution of e^{-t} and $\sin t$

$$\begin{aligned}\text{Solution : } L^{-1}(F(s)*G(s)) &= \int_0^t e^{-(t-y)} \sin y dy = e^{-t} \int_0^t e^y \sin y dy \\ &= e^{-t} \left[\frac{e^y}{1+1} (\sin y - \cos y) \right]_0^t = e^{-t} \left[\frac{e^t}{2} (\sin t - \cos t) + 1 \right] \\ &= \frac{\sin t - \cos t}{2} + \frac{e^{-t}}{2}\end{aligned}$$

45. Solve using Laplace transform $\frac{dy}{dt} + y = e^{-t}$, given that $y(0) = 0$

Solution :

Taking Laplace on Both sides, we get

$$\begin{aligned}L(Dy+y) &= L(e^{-t}) \Rightarrow sL(y) - y(0) + L(y) = \frac{1}{s+1} \\ L(y)(s+1) &= \frac{1}{s+1} \Rightarrow L(y) = \frac{1}{(s+1)^2} \Rightarrow L = \frac{1}{(s+1)^2} = e^{-t}t\end{aligned}$$

46. Find $L^{-1}\left(\frac{1+e^{-s}}{s}\right)$

$$\text{Solution : } L^{-1}\left(\frac{1+e^{-s}}{s}\right) = L^{-1}\left(\frac{1}{s}\right) + L^{-1}\left(\frac{1}{s}e^{-s}\right) = 1 + u(t-1)$$

47. If $L(f(t)) = \frac{s+2}{s^2+4}$ find the value of $\int_0^\infty f(t)dt$

$$\text{Solution : } \int_0^\infty \frac{s+2}{s^2+4} dt = \left\{ L\left(\frac{s+2}{s^2+4}\right) \right\}_{s=0} = (\cos 2t + \sin 2t)_{s=0} = 1$$

48. Find Inverse Laplace transform of $\log\left(\frac{s+1}{s-1}\right)$

Solution :

$$\begin{aligned}L^{-1}\left[\log\left(\frac{s+1}{s-1}\right)\right] &= -\frac{1}{t}L^{-1}\left[\frac{d}{ds}\log\left(\frac{s+1}{s-1}\right)\right] = -\frac{1}{t}L^{-1}\left[\frac{d}{ds}\{\log(s+1) - \log(s-1)\}\right] \\ &= -\frac{1}{t}L^{-1}\left[\frac{1}{s+1} - \frac{1}{s-1}\right] = -\frac{1}{t}[e^{-t} - e^t] = \frac{2\sinh t}{t}\end{aligned}$$

49. Find Inverse Laplace transform of $\log\left(\frac{s-5}{s^2+9}\right)$

$$\begin{aligned}\text{Solution : } L^{-1}\left\{\log\left(\frac{s-5}{s^2+9}\right)\right\} &= -\frac{1}{t}L^{-1}\left[\frac{d}{ds}\{\log(s-5) - \log(s^2+9)\}\right] \\ &= -\frac{1}{t}L^{-1}\left[\frac{1}{s-5} - \frac{2s}{s^2+9}\right] = \frac{1}{t}[2\cos 3t - e^{5t}]\end{aligned}$$

50. Find Inverse Laplace transform of $\log\left(\frac{s^2+a^2}{s^2+b^2}\right)$

Solution :

$$\begin{aligned}L\left\{\log\left(\frac{s^2+a^2}{s^2+b^2}\right)\right\} &= -\frac{1}{t}L^{-1}\left[\frac{d}{ds}\{\log(s^2+a^2) - \log(s^2+b^2)\}\right] \\ &= -\frac{1}{t}L^{-1}\left[\frac{2s}{s^2+a^2} - \frac{2s}{s^2+b^2}\right] = \frac{2(\cos bt - \cos at)}{t}\end{aligned}$$

Two Mark Questions

UNIT V

1. State Fourier integral theorem.

Sol. If $f(x)$ is piecewise continuous derivative and absolutely integrable in $(-\infty, \infty)$ then

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e^{is(x-t)} dt ds \quad (\text{or}) \quad f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos[s(x-t)] dt ds$$

2. Define Fourier transform pair.

Sol. Fourier transform of $f(x)$ is

$$F(s) = F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

Its Inverse Fourier transform is

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F[f(x)] e^{-isx} ds = F^{-1}[F(s)]$$

3. Define Fourier cosine transform pair.

Sol. Fourier cosine transform of $f(x)$ is

$$F_c(s) = F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx dx$$

Its Inverse Fourier cosine transform is

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_c[f(x)] \cos sx ds$$

4. Define Fourier sine transform pair.

Sol. Fourier sine transform of $f(x)$ is

$$F_s(s) = F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx dx$$

Its Inverse Fourier sine transform is

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s[f(x)] \sin sx ds$$

5. State Parseval's identity for Fourier transform.

Sol. If $F(s)$ is the Fourier transform of $f(x)$ then

$$\int_{-\infty}^{\infty} |F(s)|^2 ds = \int_{-\infty}^{\infty} |f(x)|^2 dx$$

6. State Parseval's identity for Fourier sine and cosine transform.

Sol. If $F_s(s)$ and $F_c(s)$ are the Fourier sine and Fourier cosine transform of $f(x)$ respectively then

$$\int_0^{\infty} [F_s(s)]^2 ds = \int_0^{\infty} [f(x)]^2 dx \quad \text{and} \quad \int_0^{\infty} [F_c(s)]^2 ds = \int_0^{\infty} [f(x)]^2 dx$$

7. Define the convolution of two functions for Fourier transform.

Sol. The convolution of two functions $f(x)$ and $g(x)$ is defined by

$$(f * g)(x) = f(x) * g(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) g(x-t) dt$$

8. State convolution theorem

Sol. If $F[f(x)] = F(s)$ and $F[g(x)] = G(s)$ then $F[f(x) * g(x)] = F(s).G(s)$

9. Solve the integral equation $\int_0^{\infty} f(x) \cos \lambda x \, dx = e^{-\lambda}$

Sol. Given $\int_0^{\infty} f(x) \cos \lambda x \, dx = e^{-\lambda}$

$$\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos \lambda x \, dx = \sqrt{\frac{2}{\pi}} e^{-\lambda}$$

$$F_c[f(x)] = \sqrt{\frac{2}{\pi}} e^{-\lambda}$$

$$f(x) = F_c^{-1} \left[\sqrt{\frac{2}{\pi}} e^{-\lambda} \right]$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sqrt{\frac{2}{\pi}} e^{-\lambda} \cos \lambda x \, d\lambda$$

$$= \frac{2}{\pi} \int_0^{\infty} e^{-\lambda} \cos \lambda x \, d\lambda$$

$$= \frac{2}{\pi} \left[\frac{e^{-\lambda}}{1+x^2} (-\cos \lambda x + \lambda \sin \lambda x) \right]_0^{\infty}$$

$$= \frac{2}{\pi} \left[\{0\} - \left\{ \frac{1}{1+x^2} (-1+0) \right\} \right]$$

$$(i.e.) f(x) = \frac{2}{\pi} \frac{1}{1+x^2}$$

10. Find $f(x)$ if its sine transform is e^{-as}

Sol. The inverse Fourier sine transform is given by

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s[f(x)] \sin sx \, ds$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-as} \sin sx \, ds$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-as}}{a^2+x^2} (-a \sin sx - x \cos sx) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[\{0\} - \left\{ \frac{1}{a^2+x^2} (0-x) \right\} \right]$$

$$= \sqrt{\frac{2}{\pi}} \frac{x}{x^2+a^2}$$

11. State the Fourier transform of the derivatives of a function.

Sol. $F[f'(x)] = (-is)F(s)$

$$F[f''(x)] = (-is)^2 F(s)$$

$$F[f'''(x)] = (-is)^3 F(s)$$

$$\text{In general, } F[f^{(n)}(x)] = (-is)^n F(s)$$

12. Find $f(x)$ if its cosine transform is $f_c(p) = \begin{cases} \frac{1}{\sqrt{2\pi}} \left(a - \frac{s}{2}\right), & s < 2a \\ 0, & s \geq 2a \end{cases}$

Sol. The inverse Fourier cosine transform is given by

$$\begin{aligned} f(x) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_c[f(x)] \cos sx \, ds \\ &= \sqrt{\frac{2}{\pi}} \left[\int_0^{2a} \frac{1}{\sqrt{2\pi}} \left(a - \frac{s}{2}\right) \cos sx \, ds + \int_{2a}^{\infty} 0 \, ds \right] \\ &= \frac{1}{\pi} \left[\left(a - \frac{s}{2}\right) \left(\frac{\sin sx}{x}\right) - \left(-\frac{1}{2}\right) \left(\frac{-\cos sx}{x^2}\right) \right]_0^{2a} \\ &= \frac{1}{\pi} \left[\left\{0 - \frac{\cos 2ax}{2x^2}\right\} - \left\{0 - \frac{1}{2x^2}\right\} \right] \\ &= \frac{1}{\pi x^2} \frac{1 - \cos 2ax}{2} \\ &= \frac{\sin^2 ax}{\pi x^2} \end{aligned}$$

13. Find the sine transform of $\frac{1}{x}$

Sol. $F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx \, dx$

$$\begin{aligned} F_s\left[\frac{1}{x}\right] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{1}{x} \sin sx \, dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{s}{t} \sin t \frac{dt}{s} \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{\sin t}{t} \, dt \\ &= \sqrt{\frac{2}{\pi}} \frac{\pi}{2} \\ &= \sqrt{\frac{\pi}{2}} \end{aligned}$$

Put $sx = t$
 $s \, dx = dt$

$\int_0^{\infty} \frac{\sin t}{t} \, dt = \frac{\pi}{2}$

14. Prove that $F[af(x) + bg(x)] = aF(s) + bG(s)$ [Linearity property on Fourier transform]

Sol. We have $F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} \, dx = F(s)$

$$\begin{aligned} F[af(x) + bg(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} [af(x) + bg(x)] e^{isx} \, dx \\ &= a \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} \, dx + b \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{isx} \, dx \\ &= aF(s) + bG(s) \end{aligned}$$

15. Prove (i) $F_c[af(x) + bg(x)] = aF_c(s) + bG_c(s)$ [Linear property on Fourier cosine transform]

(ii) $F_s[af(x) + bg(x)] = aF_s(s) + bG_s(s)$ [Linear property on Fourier sine transform]

Sol. (i) $F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx = F_c(s)$

$$\begin{aligned} F_c[af(x) + bg(x)] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} [af(x) + bg(x)] \cos sx \, dx \\ &= a \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx + b \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(x) \cos sx \, dx \\ &= a F_c(s) + b G_c(s) \end{aligned}$$

(ii) $F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx \, dx = F_s(s)$

$$\begin{aligned} F_s[af(x) + bg(x)] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} [af(x) + bg(x)] \sin sx \, dx \\ &= a \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx \, dx + b \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(x) \sin sx \, dx \\ &= a F_s(s) + b G_s(s) \end{aligned}$$

16. Prove that $F[f(x-a)] = e^{ias} F(s)$ [Time shifting property]

Sol. We have $F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} \, dx = F(s)$

$$\begin{aligned} F[f(x-a)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-a) e^{isx} \, dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{is(t+a)} \, dt \\ &= e^{ias} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{ist} \, dt \\ &= e^{ias} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} \, dx \\ &= e^{ias} F(s) \end{aligned}$$

Put $x - a = t$
 $dx = dt$

17. Prove that $F[e^{iax} f(x)] = F(s+a)$ [Frequency shifting property]

Sol. We have $F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} \, dx = F(s)$

$$\begin{aligned} F[e^{iax} f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iax} f(x) e^{isx} \, dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{i(s+a)x} \, dx \\ &= F(s+a) \end{aligned}$$

18. Prove that (i) $F[f(ax)] = \frac{1}{a} F\left(\frac{s}{a}\right)$, $a > 0$ [Change of scale property]

$$(ii) \quad F_s[f(ax)] = \frac{1}{a} F_s\left(\frac{s}{a}\right)$$

$$(iii) \quad F_c[f(ax)] = \frac{1}{a} F_c\left(\frac{s}{a}\right)$$

Sol. (i) We have $F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx = F(s)$

$$\begin{aligned} F[f(ax)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(ax) e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{is \frac{t}{a}} \frac{dt}{a} \\ &= \frac{1}{a} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i \frac{s}{a} t} dt \\ &= \frac{1}{a} F\left(\frac{s}{a}\right) \end{aligned}$$

Put $ax = t$
 $a dx = dt$

(ii) We have $F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx dx = F_s(s)$

$$\begin{aligned} F_s[f(ax)] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(ax) \sin sx dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin\left(\frac{st}{a}\right) \frac{dt}{a} \\ &= \frac{1}{a} \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin\left(\frac{s}{a} t\right) dt \\ &= \frac{1}{a} F_s\left(\frac{s}{a}\right) \end{aligned}$$

Put $ax = t$
 $a dx = dt$

(iii) We have $F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx dx = F_c(s)$

$$\begin{aligned} F_c[f(ax)] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(ax) \cos sx dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos\left(\frac{st}{a}\right) \frac{dt}{a} \\ &= \frac{1}{a} \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos\left(\frac{s}{a} t\right) dt \\ &= \frac{1}{a} F_c\left(\frac{s}{a}\right) \end{aligned}$$

Put $ax = t$
 $a dx = dt$

19. If $\bar{f}(\lambda)$ is the Fourier transform of $f(x)$, find the Fourier transform of $f(x-a)$ and $f(ax)$.

Sol. $F[f(x-a)] = e^{ia\lambda} \bar{f}(\lambda)$ [see the solution in problem 16 & 18(i)]

$$\text{and } F[f(ax)] = \frac{1}{a} \bar{f}\left(\frac{\lambda}{a}\right)$$

20. Prove that [Modulation property]

$$\begin{aligned}
 (i) \quad F[f(x)\cos ax] &= \frac{1}{2}[F(s+a) + F(s-a)] & (ii) \quad F_s[f(x)\cos ax] &= \frac{1}{2}[F_s(s+a) + F_s(s-a)] \\
 (iii) \quad F_s[f(x)\sin ax] &= \frac{1}{2}[F_c(s-a) - F_c(s+a)] & (iv) \quad F_c[f(x)\cos ax] &= \frac{1}{2}[F_c(s+a) + F_c(s-a)] \\
 (v) \quad F_c[f(x)\sin ax] &= \frac{1}{2}[F_s(a+s) + F_s(a-s)]
 \end{aligned}$$

Sol. (i) We have $F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx = F(s)$

$$\begin{aligned}
 F[f(x)\cos ax] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\cos ax e^{isx} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \left(\frac{e^{iax} + e^{-iax}}{2} \right) e^{isx} dx \\
 &= \frac{1}{2} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{i(s+a)x} dx + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{i(s-a)x} dx \right] \\
 &= \frac{1}{2} [F(s+a) + F(s-a)]
 \end{aligned}$$

(ii) We have $F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx dx = F_s(s)$

$$\begin{aligned}
 F_s[f(x)\cos ax] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x)\cos ax \sin sx dx & 2\sin A \cos B &= \sin(A+B) + \sin(A-B) \\
 &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \frac{1}{2} [\sin(s+a)x + \sin(s-a)x] dx \\
 &= \frac{1}{2} \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin(s+a)x dx + \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin(s-a)x dx \right] \\
 &= \frac{1}{2} [F_s(s+a) + F_s(s-a)]
 \end{aligned}$$

(iii) We have $F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx dx = F_s(s)$

$$\begin{aligned}
 F_s[f(x)\sin ax] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x)\sin ax \sin sx dx & 2\sin A \sin B &= \cos(A-B) - \cos(A+B) \\
 &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \frac{1}{2} [\cos(s-a)x - \cos(s+a)x] dx \\
 &= \frac{1}{2} \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos(s-a)x dx - \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos(s+a)x dx \right] \\
 &= \frac{1}{2} [F_c(s-a) - F_c(s+a)]
 \end{aligned}$$

(iv) We have $F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx dx = F_c(s)$

$$F_c[f(x)\cos ax] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x)\cos ax \cos sx dx \quad 2\cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$\begin{aligned}
&= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \frac{1}{2} [\cos(s+a)x + \cos(s-a)x] dx \\
&= \frac{1}{2} \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos(s+a)x dx + \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos(s-a)x dx \right] \\
&= \frac{1}{2} [F_c(s+a) + F_c(s-a)]
\end{aligned}$$

(v) We have $F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx dx = F_c(s)$

$$\begin{aligned}
F_c[f(x) \sin ax] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin ax \cos sx dx \quad \boxed{2 \sin A \cos B = \sin(A+B) + \sin(A-B)} \\
&= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \frac{1}{2} [\sin(a+s)x + \sin(a-s)x] dx \\
&= \frac{1}{2} \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin(a+s)x dx + \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin(a-s)x dx \right] \\
&= \frac{1}{2} [F_s(a+s) + F_s(a-s)]
\end{aligned}$$

21. Prove that (i) $F[f(-x)] = F(-s)$ (ii) $F[\overline{f(-x)}] = \overline{F(s)}$ (iii) $F[\overline{f(x)}] = \overline{F(-s)}$

Sol. (i) We have $F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx = F(s)$

$$\begin{aligned}
F[f(-x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(-x) e^{isx} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{\infty}^{-\infty} f(t) e^{-ist} (-dt) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-ist} dt \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i(-s)t} dt \\
&= F(-s)
\end{aligned}$$

Put $-x = t$
 $-dx = dt$

(ii) We have $F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$

$$\begin{aligned}
\overline{F(s)} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \overline{f(x)} e^{-isx} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{\infty}^{-\infty} \overline{f(-t)} e^{ist} (-dt) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \overline{f(-t)} e^{ist} dt \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \overline{f(-x)} e^{isx} dx \\
&= F[\overline{f(-x)}]
\end{aligned}$$

Put $-x = t$
 $-dx = dt$

(iii) We have $F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$

$$F(-s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-isx} dx$$

$$\overline{F(-s)} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \overline{f(x)} e^{isx} dx$$

$$= F[\overline{f(x)}]$$

22. Prove that (i) $\int_0^{\infty} F_c(s) G_c(s) ds = \int_0^{\infty} f(x) g(x) dx$

(ii) $\int_0^{\infty} F_s(s) G_s(s) ds = \int_0^{\infty} f(x) g(x) dx$

Sol. (i) $\int_0^{\infty} F_c(s) G_c(s) ds = \int_0^{\infty} F_c(s) \left\{ \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(x) \cos sx dx \right\} ds$

$$= \int_0^{\infty} g(x) \left\{ \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_c(s) \cos sx ds \right\} dx$$

$$= \int_0^{\infty} g(x) f(x) dx$$

$$= \int_0^{\infty} f(x) g(x) dx$$

(ii) $\int_0^{\infty} F_s(s) G_s(s) ds = \int_0^{\infty} F_s(s) \left\{ \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(x) \sin sx dx \right\} ds$

$$= \int_0^{\infty} g(x) \left\{ \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s(s) \sin sx ds \right\} dx$$

$$= \int_0^{\infty} g(x) f(x) dx$$

$$= \int_0^{\infty} f(x) g(x) dx$$

23. Give an example for self-reciprocal under Fourier transform.

Sol. $e^{-\frac{x^2}{2}}$ is self-reciprocal under Fourier transform.

24. Give an example for self-reciprocal under Fourier cosine transform.

Sol. $e^{-\frac{x^2}{2}}$ is self-reciprocal under Fourier cosine transform.

25. Give an example for self-reciprocal under both Fourier sine and cosine transform.

Sol. $\frac{1}{\sqrt{x}}$ is self-reciprocal under both Fourier sine and cosine transform.
